

1 The theta term and Axion Electrodynamics

In relativistic notation, the Maxwell action for electromagnetism takes a wonderfully compact form:

$$S_{Max} = \int d^4x -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} = \int d^4x \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) \quad (1)$$

Here:

$$\left\{ \begin{array}{l} F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \\ = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix} \\ E_i = F_{0i} \\ F_{ij} = -\epsilon_{ijk}B_k \end{array} \right. \quad (2)$$

One reason that the Maxwell action is so simple is that there is very little else we can write down that is both gauge invariant and Lorentz invariant. There is, however one term that we can add to the Maxwell action is also both gauge invariant and Lorentz invariant, which we call it theta term:

$$S_\theta = \frac{\theta e^2}{4\pi^2} \int d^4x \frac{1}{4} \star F^{\mu\nu} F_{\mu\nu} = -\frac{\theta e^2}{4\pi^2} \int d^4x \mathbf{E} \cdot \mathbf{B} \quad (3)$$

which $\star F^{\mu\nu}$ is the dual tensor:

$$\begin{aligned} \star F^{\mu\nu} &= \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} \\ &= \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z & -E_y \\ B_y & -E_z & 0 & E_x \\ B_z & E_y & -E_x & 0 \end{pmatrix} \end{aligned} \quad (4)$$

and θ is a constant parameter.

However, the theta term is simple to check that it can be written as a total derivative:

$$\begin{aligned} \int d^4x \frac{1}{4} \star F^{\mu\nu} F_{\mu\nu} &= \frac{1}{2} \int d^4x \cdot \frac{1}{4} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} F_{\mu\nu} \\ &= \frac{1}{8} \int d^4x \cdot \epsilon^{\mu\nu\rho\sigma} (\partial_\rho A_\sigma - \partial_\sigma A_\rho) (\partial_\mu A_\nu - \partial_\nu A_\mu) \\ &= \frac{1}{2} \int d^4x \cdot \epsilon^{\mu\nu\rho\sigma} (\partial_\mu A_\nu \partial_\rho A_\sigma) \end{aligned} \quad (5)$$

Now let's proof a useful equation:

$$\epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma = 0 \quad (6)$$

Because:

$$\begin{aligned}
 \epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma &= \epsilon^{\rho\nu\mu\sigma} A_\nu \partial_\rho \partial_\mu A_\sigma \\
 &= \epsilon^{\rho\nu\mu\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma \\
 &= -\epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma
 \end{aligned} \tag{7}$$

So we can get:

$$2\epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma = 0 \Rightarrow \epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\mu \partial_\rho A_\sigma = 0 \tag{8}$$

So the quadratic part of theta term become:

$$\begin{aligned}
 \int d^4x \frac{1}{4} \star F^{\mu\nu} F_{\mu\nu} &= \frac{1}{2} \int d^4x \epsilon^{\mu\nu\rho\sigma} (\partial_\mu A_\nu \partial_\rho A_\sigma + A_\nu \partial_\mu \partial_\rho A_\sigma) \\
 &= \frac{1}{2} \int d^4x \epsilon^{\mu\nu\rho\sigma} \partial_\mu (A_\nu \partial_\rho A_\sigma)
 \end{aligned} \tag{9}$$

The total derivative form of theta term is:

$$S_\theta = \frac{\theta e^2}{8\pi^2} \int d^4x \partial_\mu (\epsilon^{\mu\nu\rho\sigma} A_\nu \partial_\rho A_\sigma) \tag{10}$$

We say that the theta term is topological. It depends only on boundary information. However, when deriving using the principle of least action to derive the field's equation of motion, the values of field are fixed on the infinite boundary. Therefore, the upshot is that the theta term does not change the equations of motion and, it would seem, can have little effect on the physics.

But under some situations that involve subtle interplay between quantum mechanics and topology, there are a number of interesting phenomena of physics which are led by theta term.

Also, we can look at the situations where θ affects the dynamics classically. This occurs when θ is not constant, but instead varies in space and time.

$$\theta = \theta(x, t) \tag{11}$$

So we can write down the axion electrodynamics action:

$$S = \int d^4x \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{e^2}{16\pi^2} \theta(x, t) \star F^{\mu\nu} F_{\mu\nu} \right) \tag{12}$$

We use Euler-Lagrange equations:

$$\frac{\partial \mathcal{L}}{\partial A_\mu} - \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu A_\mu)} \right) = 0 \tag{13}$$

to get the equations of axion electrodynamics

Because $\mathcal{L}$ doesn't depend on A_μ , so the equations of motion are:

$$\frac{\partial \mathcal{L}}{\partial A_\mu} = 0 \Rightarrow \partial_\nu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\nu A_\mu)} \right] = 0 \tag{14}$$

and the $\mathcal{L}$ of axion electrodynamics is ($\alpha = \frac{e^2}{4\pi}$):

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{Max} + \mathcal{L}_\theta \\ &= -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \frac{\alpha}{4\pi}\theta(\mathbf{x}, t)\star F^{\mu\nu}F_{\mu\nu}\end{aligned}\quad (15)$$

First, let's see $\mathcal{L}_\theta$:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial(\partial_\nu A_\mu)} &= \frac{\partial}{\partial(\partial_\nu A_\mu)} \cdot \frac{\alpha}{4\pi}\theta(\mathbf{x}, t) \cdot \frac{1}{2}\epsilon^{\alpha\beta\rho\sigma}F_{\alpha\beta}F_{\rho\sigma} \\ &= \frac{\alpha}{8\pi}\left\{\left[\frac{\partial}{\partial(\partial_\nu A_\mu)}\theta(\mathbf{x}, t)\right] \cdot \epsilon^{\alpha\beta\rho\sigma}F_{\alpha\beta}F_{\rho\sigma} + \theta(\mathbf{x}, t)\left[\frac{\partial\epsilon^{\alpha\beta\rho\sigma}F_{\alpha\beta}F_{\rho\sigma}}{\partial(\partial_\nu A_\mu)}\right]\right\}\end{aligned}\quad (16)$$

let's see the first term:

$$\epsilon^{\alpha\beta\rho\sigma}F_{\alpha\beta}F_{\rho\sigma}\frac{\partial}{\partial(\partial_\nu A_\mu)}\theta(\mathbf{x}, t) = 0 \quad (17)$$

then let's see the second term:

$$\begin{aligned}\theta(\mathbf{x}, t) \cdot \frac{\partial}{\partial(\partial_\nu A_\mu)} \cdot \epsilon^{\alpha\beta\rho\sigma}F_{\alpha\beta}F_{\rho\sigma} &= 4\theta(\mathbf{x}, t)\left(\frac{\partial}{\partial(\partial_\nu A_\mu)}\partial_\alpha A_\beta\partial_\rho A_\sigma\right)\epsilon^{\alpha\beta\rho\sigma} \\ &= 4\theta(\mathbf{x}, t)(\delta_\alpha^\nu\delta_\beta^\mu\partial_\rho A_\sigma + \delta_\rho^\nu\delta_\sigma^\mu\partial_\alpha A_\beta) \cdot \epsilon^{\alpha\beta\rho\sigma} \\ &= 4\theta(\mathbf{x}, t) \cdot \epsilon^{\nu\mu\rho\sigma}(\partial_\rho A_\sigma - \partial_\sigma A_\rho) \\ &= 8\theta(\mathbf{x}, t)\star F^{\nu\mu}\end{aligned}\quad (18)$$

next, we can get:

$$\partial_\nu[\theta(\mathbf{x}, t)\star F^{\nu\mu}] = \star F^{\nu\mu}\partial_\nu\theta(\mathbf{x}, t) + \theta(\mathbf{x}, t) \cdot \partial_\nu\star F^{\nu\mu} \quad (19)$$

using the Bianchi identities:

$$\partial_\mu\star F^{\mu\nu} = 0 \quad (20)$$

so we can get:

$$\partial_\nu\theta(\mathbf{x}, t)\star F^{\nu\mu} = \star F^{\nu\mu}\partial_\nu\theta(\mathbf{x}, t) \quad (21)$$

Finally, we can get:

$$\partial_\nu\left[\frac{\partial \mathcal{L}_\theta}{\partial(\partial_\nu A_\mu)}\right] = \frac{\alpha}{\pi}\star F^{\nu\mu} \cdot \partial_\nu\theta(\mathbf{x}, t) \quad (22)$$

And the Lagrangian of Maxwell part:

$$\partial_\nu\left[\frac{\partial \mathcal{L}_{Max}}{\partial(\partial_\nu A_\mu)}\right] = -\partial_\nu F^{\nu\mu} \quad (23)$$

So the equations of axion electrodynamics is:

$$\begin{cases} \partial_\nu F^{\nu\mu} = \frac{\alpha}{\pi}\star F^{\nu\mu} \cdot \partial_\nu\theta(\mathbf{x}, t) \\ \partial_\mu\star F^{\mu\nu} = 0 \end{cases} \quad (24)$$

In the end, the deformed Maxwell equations which are the equations of axion electrodynamics are:

$$\begin{cases} \nabla \cdot \mathbf{E} = \frac{\alpha}{\pi} (\nabla \theta) \cdot \mathbf{B} \\ \nabla \times \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t} - \frac{\alpha}{\pi} (\dot{\theta} \mathbf{B} + \nabla \theta \times \mathbf{E}) \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \end{cases} \quad (25)$$

The first equation tells us that in regions of space where θ varies, a magnetic field $\mathbf{B}$ acts like an electric charge density. The second equation tells us that the combination $(\dot{\theta} \mathbf{B} + \nabla \theta \times \mathbf{E})$ acts like a current density

1.1 The periodicity of theta and its values under Parity or Time - Reversal Symmetry

In quantum theory, θ is a periodic variable: it lies in the range:

$$\theta \in [0, 2\pi) \quad (26)$$

After imposing appropriate boundary conditions, S_θ can only take values of the form:

$$S_\theta = \theta N \quad \text{with } N \in \mathbb{Z} \quad (27)$$

This means that the theta angle contributes to the partition function as

$$e^{iS_\theta} = e^{iN\theta} \quad (28)$$

To show that S_θ must take the form, we consider a compact Euclidean spacetime which we take to T^4 and we take each of the circles in the torus to have radii R .

We consider easy case:

$$\begin{cases} \mathbf{E} = (0, 0, E_z), \quad E_z = F_{03} = \partial_0 A_3 - \partial_3 A_0 \\ \mathbf{B} = (0, 0, B_z), \quad B_z = F_{21} = \partial_2 A_1 - \partial_1 A_2 \end{cases} \quad (29)$$

The integral of S_θ is:

$$I = \int_{T^4} d^4x E_z \cdot B_z = \int_{T^4} dx^0 dx^3 \cdot E_z \cdot \int_{T^4} dx^1 dx^2 \cdot B_z \quad (30)$$

The gauge field A_μ must be well defined on the underlying torus, which will put restrictions on the allowed values of E_z and B_z . So the integral can't take any value.

Now let consider the restrictions of A_μ if A_μ be defined on torus. When a direction of space, say x^1 , is periodic with radius R , the physics state in original point ($x^1 = 0$) must be identical to the final physics state:

$$\text{state}[A_1(0)] = \text{state}[A_1(2\pi R)] \quad (31)$$

Because $B_z = \partial_2 A_1$, $(A_2 = 0) = \text{constant}$, so $A_1 = B_z x^2 + C$. The simplest case is $A_1 = 0$ but this case is trivial. So we should find a gauge transformations to connect $A(0)$ and $A(2\pi R)$, which will make two different values of A_1 correspond to the same physics state.

$$A_1(2\pi R) = A_1(0) + G \quad (32)$$

let's find the gauge transformation that we are interested:

$$A_1 \rightarrow A_1 + \partial_1 \chi \quad (33)$$

We do not insist that $\chi(x)$ is single valued. Instead, we require only that $e^{ie\chi}$ is single-valued, since this is what acts on the wavefunction. So χ can have the simple form followed:

$$\chi = \frac{nx^1}{eR} \quad \text{with } n \in \mathbb{Z} \quad (34)$$

Because in the same point $(2\pi NR + X)$, $e^{ie\chi}$ is the same state:

There are sometimes called large gauge transformations, a name which reflects the fact that they cannot be continuously deformed to the identity. Under such a gauge transformation, we can see that:

$$A_1 \rightarrow A_1 + \partial_1 \chi = A_1 + \frac{n}{eR} \quad (35)$$

the two different values of A_1 correspond the same physics state. From this transformation, we can determine the expression of A_1 ,

$$\begin{aligned} A_1(0) &= A_1(2\pi R) + \frac{n}{eR} \\ \Rightarrow A_1 &= \frac{nx^2}{2\pi eR^2} \end{aligned} \quad (36)$$

And we can write down the expression of B_z :

$$B_z = \partial_2 A_1 = \frac{n}{2\pi eR^2} \quad (37)$$

So the integral of magnetic field is:

$$\int_{T^2} dx^1 dx^2 B_z = \frac{2\pi n}{e} \quad \text{with } n \in \mathbb{Z} \quad (38)$$

We can also apply exactly the same argument to the electric field

$$E_z = \partial_0 A_3 \quad (A_0 = 0) \quad (39)$$

So the integral of electric field is:

$$\int_{T^2} dx^0 dx^3 E_z = \frac{2\pi n'}{e} \quad \text{with } n' \in \mathbb{Z} \quad (40)$$

Armed with two integrals above, the S_θ becomes:

$$\begin{aligned}
 S_\theta &= -\frac{\theta e^2}{4\pi^2} \int_{T^4} d^4x \mathbf{E} \cdot \mathbf{B} \\
 &= -\frac{\theta e^2}{4\pi^2} \frac{4\pi^2}{e^2} \cdot nn' \quad (\text{let } N = -nn') \\
 &= N\theta \quad \text{with } N \in \mathbb{Z}
 \end{aligned} \tag{41}$$

It's pretty straightforward to generalise it to non-constant $\mathbf{E}$ and $\mathbf{B}$ fields:

$$\begin{aligned}
 \int_{T^2} dx^1 dx^2 B_z &= \int_{T^2} dx^1 dx^2 \cdot \partial_2 A_1 \\
 &= 2\pi R [A_1(2\pi R) - A_1(0)] \\
 &= \frac{2\pi n}{e}
 \end{aligned} \tag{42}$$

So we can see that θ is a periodic variable in quantum theory.

Although the theta term is gauge invariant and Lorentz invariant, it is not invariant under certain discrete symmetries, and does not preserve the same symmetries as the Maxwell term. These discrete symmetries are parity $\mathcal{P}$ and time reversal $\mathcal{T}$.

$$\begin{cases} \mathcal{P} : x \rightarrow -x \\ \mathcal{T} : t \rightarrow -t \end{cases} \tag{43}$$

If we act on both $\mathbf{E}$ and $\mathbf{B}$

$$\begin{cases} \mathcal{P} : \mathbf{E}(x, t) \rightarrow -\mathbf{E}(-x, t) \text{ and } \mathcal{P} : \mathbf{B}(x, t) \rightarrow \mathbf{B}(-x, t) \\ \mathcal{T} : \mathbf{E}(x, t) \rightarrow \mathbf{E}(x, -t) \text{ and } \mathcal{T} : \mathbf{B}(x, t) \rightarrow -\mathbf{B}(x, -t) \end{cases} \tag{44}$$

So $\mathbf{E}$ is odd under parity and even under time reversal. $\mathbf{B}$ is even under parity and odd under time reversal.

This means that, the theta term $S_\theta \propto \int d^4x \mathbf{E} \cdot \mathbf{B}$ breaks both parity and time reversal invariance, which means $S_\theta \rightarrow -S_\theta$ under $\mathcal{P}$ and $\mathcal{T}$. But because of the periodicity of θ , there are two exceptions:

$$S_\theta = -S_\theta + 2\pi N \tag{45}$$

In order to make the equation hold for any integer values of N , θ must equal to 0 or π ($\theta \in [0, 2\pi)$).

So we can know that when $\theta = 0$ or $\theta = \pi$, the theory is invariant under $\mathcal{P}$ and $\mathcal{T}$ symmetries.

In subsequent sections, we can see that the topological insulator said to have a $\mathbb{Z}_2$ classification has $\theta = \pi$ because these materials are defined to be time-reversal invariant. Most materials have $\theta = 0$. But some materials have a band structure which is twisted in a particular way, which results in $\theta = \pi$.

1.2 Continuity Conditions

Let's consider a equivalent way to describing the physics above. We introduce the electric displacement:

$$\mathbf{D} = \mathbf{E} - \frac{\alpha\theta}{\pi} \cdot \mathbf{B} \quad (46)$$

because the divergence of $\mathbf{B}$ equal to zero, so:

$$\nabla \cdot \mathbf{D} = \nabla \cdot \mathbf{E} - \frac{\alpha}{\pi} (\nabla \cdot \theta) \mathbf{B} = 0. \quad (47)$$

Comparing to the expression of $\mathbf{D}$, we see that a magnetic field $\mathbf{B}$ act like polarisation in a topological insulator. When the θ varies, we have a varying polarisation resulting in bound charge. Similarly, we define the magnetising field

$$\mathbf{H} = \mathbf{B} + \frac{\alpha}{\pi} \theta \cdot \mathbf{E} \quad (48)$$

Using the relationship between $\mathbf{B}$ and $\mathbf{E}$:

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0 \quad (49)$$

so we can get:

$$\begin{aligned} \nabla \times \mathbf{H} &= \nabla \times \mathbf{B} + \frac{\alpha}{\pi} [(\nabla \theta) \times \mathbf{E} + \theta (\nabla \times \mathbf{E})] \\ &= \frac{\partial \mathbf{D}}{\partial t} + \frac{\alpha}{\pi} \theta \cdot \left(\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} \right) \\ &= \frac{\partial \mathbf{D}}{\partial t} \end{aligned} \quad (50)$$

In topological insulator, $\mathbf{E}$ act like magnetisation. When θ varies, we get a varying magnetisation which results in bound currents.

With these definitions, the equations of axion electrodynamics take the usual form of Maxwell equations in matter:

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{D} = 0 \\ \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = 0 \\ \nabla \cdot \mathbf{B} = 0 \\ \frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0 \end{array} \right. \quad (51)$$

This means that if we introduce the normal vector to the surface $\hat{n}$, we have the boundary conditions:

$$\begin{cases} \hat{n} \cdot \Delta \mathbf{D} = 0 \\ \hat{n} \times \Delta \mathbf{E} = 0 \\ \hat{n} \cdot \Delta \mathbf{B} = 0 \\ \hat{n} \times \Delta \mathbf{H} = 0 \end{cases} \quad (52)$$

1.3 The Witten effect

The effect of the θ term is to endow the magnetic monopole with an electric charge, which is known as the Witten effect

Let's take a magnetic monopole with magnetic charge g and place it inside a vacuum ($\theta = 0$). Then, we surround this with a medium that has $\theta = \Theta \neq 0$

When the magnetic field crosses the interface where θ changes, it will induce an electric charge:

$$\begin{aligned} \int \nabla \cdot \mathbf{E} dV &= \int \frac{\alpha}{\pi} (\nabla \cdot \theta) \cdot \mathbf{B} dV. \\ \Rightarrow Q &= \frac{\alpha}{\pi} \Theta g \end{aligned} \quad (53)$$

This result is independent of the size of the interior region whose θ is zero. So we could shrink this region down until it is infinitesimally small, and we still find that the monopole has charge q : In fact a monopole is a dyon which carries electric charge when θ not equal to zero.

When the monopole carries the minimum allowed magnetic charge, its electric charge is given by:

$$g = \frac{2\pi}{e} \Rightarrow q = \frac{e\theta}{2\pi} \quad (54)$$

In particular, if we place a magnetic monopole inside a topological insulator, it turns into a dyon which carries half the charge of the electron.

We can also consider placing a monopole in a medium whose θ will gradually increase from zero. At the end of this process, the final electric field will be:

$$\begin{aligned} \int_0^{t_0} \frac{\partial \mathbf{E}}{\partial t} dt &= \frac{\alpha}{\pi} \int_0^{t_0} \dot{\theta} \mathbf{B} \\ \Rightarrow \mathbf{E} &= \frac{\alpha}{\pi} \Theta \mathbf{B} \quad [\text{which } \theta(t_0) = \Theta] \\ \Rightarrow q &= \frac{\alpha}{\pi} \Theta g \end{aligned} \quad (55)$$

1.4 A Mirage Monopole

let's consider that we take an electric charge q and place it in the vacuum at point $\mathbf{x} = (0, 0, d)$. then we fill the region where z is less than zero with topological insulator. We can see that the

effect produced by surface charge is equivalent to placing mirror dyons at point $\mathbf{x} = (0, 0, d)$ and point $\mathbf{x}' = (0, 0, -d)$.

We assume that the mirror dyon sitting at $\mathbf{x} = (0, 0, d)$ with electric and magnetic charge (q_1, g_1) and the mirror dyon sitting at $\mathbf{x}' = (0, 0, -d)$ with electric and magnetic charge (q_2, g_2) .

Next, we can write down:

$$\begin{cases} \mathbf{E} = -\nabla\phi & (\text{because } \nabla \times \mathbf{E} = 0) \\ \mathbf{B} = -\nabla\Omega & (\text{because } \nabla \times \mathbf{B} = 0) \end{cases} \quad (56)$$

So the electric potential in the two regions is:

$$\begin{cases} \phi_+ = \frac{1}{4\pi} \left[\frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} + \frac{q_2}{\sqrt{x^2 + y^2 + (z+d)^2}} \right], & z > 0 \\ \phi_- = \frac{1}{4\pi} \frac{q + q_1}{\sqrt{x^2 + y^2 + (z-d)^2}}, & z < 0 \end{cases} \quad (57)$$

We only need to consider the electric and magnetic field in yz plane because of symmetry, where z equal to zero

$$\begin{cases} \left. \begin{aligned} E_y^+ &= -\partial_y \phi_+ = \frac{q + q_2}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ E_z^+ &= -\partial_z \phi_+ = \frac{q_2 - q}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{aligned} \right\} z = 0^+ \\ \left. \begin{aligned} E_y^- &= -\partial_y \phi_- = \frac{q + q_1}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ E_z^- &= -\partial_z \phi_- = \frac{-(q + q_1)}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{aligned} \right\} z = 0^- \end{cases} \quad (58)$$

Using the boundary conditions:

$$\hat{\mathbf{n}} \times \mathbf{E} = 0 \quad (59)$$

so we can get the equation:

$$E_y^+ = E_y^- \Rightarrow q_1 = q_2 = q' \quad (60)$$

Also, we can perform the same operation on the magnetic field; The magnetic scalar potential is:

$$\begin{cases} \Omega_+ = \frac{1}{4\pi} \cdot \frac{g_2}{\sqrt{x^2 + y^2 + (z+d)^2}}, & z > 0 \\ \Omega_- = \frac{1}{4\pi} \cdot \frac{g_1}{\sqrt{x^2 + y^2 + (z-d)^2}}, & z < 0 \end{cases} \quad (61)$$

So the magnetic field on yz plane where z equal to zero:

$$\left\{ \begin{array}{l} B_y^+ = -\partial_y \Omega_+ = \frac{g_2}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ B_z^+ = -\partial_z \Omega_+ = \frac{g_2}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{array} \right\} z = 0^+ \quad (62)$$

$$\left\{ \begin{array}{l} B_y^- = -\partial_y \Omega_- = \frac{g_1}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ B_z^- = -\partial_z \Omega_- = \frac{-g_1}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{array} \right\} z = 0^-$$

Using the boundary conditions:

$$\hat{n} \cdot \mathbf{B} = 0 \Rightarrow B_z^- = B_z^+ \Rightarrow g_2 = -g_1 = g \quad (63)$$

From the expression above, we can get the electric displacement:

$$\left\{ \begin{array}{l} D_z^+ = E_z^+ = \frac{q' - q}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ D_z^- = E_z^- - \frac{\alpha}{\pi} \theta B_z^- = \frac{-(q + q') + \alpha g}{4\pi} \cdot \frac{d}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{array} \right. \quad (64)$$

Using the boundary conditions:

$$\hat{n} \cdot \mathbf{D} = 0 \Rightarrow D_z^+ = D_z^- \Rightarrow g = \frac{-2q'}{\alpha} \quad (65)$$

The magnetising field:

$$\left\{ \begin{array}{l} H_y^+ = B_y^+ = \frac{g}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \\ H_y^- = B_y^- + \frac{\alpha}{\pi} \theta E_y^- = \frac{-g + \alpha(q + q')}{4\pi} \cdot \frac{y}{(x^2 + y^2 + d^2)^{\frac{3}{2}}} \end{array} \right. \quad (66)$$

Using the boundary condition:

$$\nabla \times \mathbf{H} = 0 \Rightarrow H_y^+ = H_y^- \Rightarrow g = \frac{\alpha(q + q')}{2} \quad (67)$$

Finally, we can get the electric and magnetic charges carried by these dyons:

$$\left\{ \begin{array}{l} q' = \frac{-\alpha^2}{4 + \alpha^2} \cdot q \\ g = \frac{2\alpha}{4 + \alpha^2} \cdot q \end{array} \right. \quad (68)$$

2 3D Topological Insulator

Let's consider the low-energy continuum model for prototypical 3D topological insulators such as Bi_2Se_3 . The bulk electronic structure of Bi_2Se_3 near the Fermi level is described by two p-orbitals $P1_z^+$ and $P2_z^-$ with $\pm$ denoting parity. Defining the basis $\{|P1_z^+, \uparrow\rangle, |P1_z^+, \downarrow\rangle, |P2_z^-, \uparrow\rangle, |P2_z^-, \downarrow\rangle\}$ and retaining the wave vector $\mathbf{k}$ up to quadratic order, the low-energy effective Hamiltonian around the Γ point is given by:

$$H_{\text{eff}}(\mathbf{k}) = \begin{bmatrix} M(\mathbf{k}) & 0 & A_1 k_z & A_2 k_- \\ 0 & M(\mathbf{k}) & A_2 k_+ & -A_1 k_z \\ A_1 k_z & A_2 k_- & -M(\mathbf{k}) & 0 \\ A_2 k_+ & -A_1 k_z & 0 & -M(\mathbf{k}) \end{bmatrix} \quad (69)$$

$$= A_2 k_x \alpha_1 + A_2 k_y \alpha_2 + A_1 k_z \alpha_3 + M(\mathbf{k}) \alpha_4$$

where $k_{\pm} = k_x \pm i k_y$ and $M(\mathbf{k}) = m_0 - B_1 k_z^2 - B_2(k_x^2 + k_y^2)$. The 4×4 matrices α_{μ} are given by the Dirac representation:

$$\alpha_j = \begin{bmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{bmatrix}, \quad \alpha_4 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (70)$$

Next, let's consider the lattice model of above Hamiltonian, which can calculate the $\mathbb{Z}_2$ invariant identifying whether a phase is topological nontrivial or trivial.

The simplest 3D lattice is the cubic lattice. We make the changes as following (ignore lattice constant a).

$$\begin{cases} k_i \rightarrow \sin k_i \\ k_i^2 \rightarrow 2(1 - \cos k_i) \end{cases} \quad (71)$$

and simplify the coefficients to obtain the isotropic lattice Hamiltonian:

$$H_{\text{eff}}(\mathbf{k}) = A(\alpha_1 \sin k_x + \alpha_2 \sin k_y + \alpha_3 \sin k_z) + [m_0 + r \sum (1 - \cos k_i)] \alpha_4 \quad (72)$$

where we define $A = A_1 = A_2$ and $r = -2B_1 = -2B_2$. Then, we use Fu-Kane method to calculate the $\mathbb{Z}_2$ topological invariant for this system:

$$(-1)^{\nu} = \prod_{i=1}^{N_{\text{TRIM}}} \text{sgn}[M(\Lambda_i)] \quad (73)$$

In this 3D cubic lattice, there are 8 TRIM (Time-Reversal Invariant Momenta which are invariant under $k_i \rightarrow -k_i$) point :

$$\begin{cases} \Gamma : (0, 0, 0) \\ X : (\pi, 0, 0) / (0, \pi, 0) / (0, 0, \pi) \\ M : (\pi, \pi, 0) / (\pi, 0, \pi) / (0, \pi, \pi) \\ R : (\pi, \pi, \pi) \end{cases} \quad (74)$$

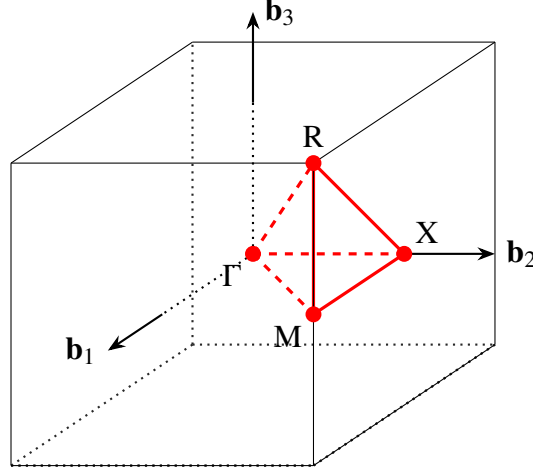


Figure 1: Highly symmetric points in the Brillouin zone of a cubic lattice

The mass term is:

$$M(\mathbf{k}) = m_0 + r \sum_{t=x,y,z} (1 - \cos k_t) \quad (75)$$

so we can calculate the mass term for each group of point:

$$\begin{cases} \Gamma : M = m_0 \\ X : M = m_0 + 2r \\ M : M = m_0 + 4r \\ R : M = m_0 + 6r \end{cases} \quad (76)$$

Now we substitute these mass terms into the $\mathbb{Z}_2$ topological invariant formula

$$\begin{aligned} (-1)^\nu &= \text{sgn}(m_0) \times \text{sgn}(m_0 + 2r)^3 \times \text{sgn}(m_0 + 4r)^3 \times \text{sgn}(m_0 + 6r) \\ &= \text{sgn}(m_0) \times \text{sgn}(m_0 + 2r) \times \text{sgn}(m_0 + 4r) \times \text{sgn}(m_0 + 6r) \\ &= \begin{cases} -1 & (0 > \frac{m_0}{r} > -2, -4 > \frac{m_0}{r} > -6) \\ 1 & (\frac{m_0}{r} > 0, -2 > \frac{m_0}{r} > -4, -6 > \frac{m_0}{r}) \end{cases} \end{aligned} \quad (77)$$

let's return to the continuum isotropic Hamiltonian to obtain the θ term. Consider the Hamiltonian around the Γ point can be simplified by ignoring the terms second-order in k_i :

$$H_{\text{TI}}(\mathbf{k}) = A\mathbf{k} \cdot \boldsymbol{\alpha} + m_0\alpha_4 \quad (78)$$

Because this Hamiltonian describes a topological insulator, m_0 is less than zero ($(-1)^\nu = \text{sgn}(m_0)$). Except for the negative mass m_0 , this is the usual Dirac Hamiltonian ($A\mathbf{k} \rightarrow c\mathbf{p}$). In the presence of an external electromagnetic vector potential $\mathbf{A}$, minimal coupling results in:

$$\mathbf{p} \rightarrow \mathbf{p} + e\mathbf{A} \quad (79)$$

where $e > 0$ being the magnitude of the electron charge. At the same time, because of the external electromagnetic scalar potential A_0 , the energy density is modified as:

$$\psi^\dagger H_0 \psi \rightarrow \psi^\dagger (H_0 - eA_0) \psi \quad (80)$$

So the action of the system in the presence of an external electromagnetic four potential $A_\mu = (A_0, -\mathbf{A})$ is written in the follow relativistic form:

$$\begin{aligned} S_{\text{TI}} &= \int dt d^3x \psi^\dagger \{i(\partial_t - ieA_0) - [H_{\text{TI}}(\mathbf{k} + e\mathbf{A})]\} \psi \\ &= \int dt d^3x \psi^\dagger [i(\partial_t - ieA_0) - (-i\nabla + e\mathbf{A}) \cdot \boldsymbol{\alpha} - m_0\alpha_4] \psi \end{aligned} \quad (81)$$

where $\psi^\dagger(\mathbf{x}, t)$ is a fermionic field representing the basis of the Hamiltonian.

Then we need to convert from the basis of $\psi^\dagger$ and α_j matrix to the relativistic form basis of $\bar{\psi}$ and γ matrices:

$$\left\{ \begin{aligned} \gamma^0 &= \alpha_4 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ \gamma^j &= \alpha_4 \alpha_j = \begin{bmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{bmatrix} \\ \gamma^5 &= i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}, \\ \bar{\psi} &= \psi^\dagger \gamma^0 \end{aligned} \right. \quad (82)$$

To introduce $\bar{\psi}$, we use the identity $(\gamma^0)^2 = \mathbb{I}_{4 \times 4}$, so the Lagrangian can be rewritten as:

$$\begin{aligned} \mathcal{L}_{\text{TI}} &= \bar{\psi} \gamma^0 [i(\partial_t - ieA_0) + i\nabla \cdot \boldsymbol{\alpha} - e\mathbf{A} \cdot \boldsymbol{\alpha} - m_0\alpha_4] \psi \\ &= \bar{\psi} [\gamma^0(i\partial_t + i\nabla \cdot \boldsymbol{\alpha}) + \gamma^0(eA_0 - e\mathbf{A} \cdot \boldsymbol{\alpha}) - m_0\gamma^0\alpha_4] \psi \\ &= \bar{\psi} [i(\gamma^0\partial_t + \boldsymbol{\gamma} \cdot \boldsymbol{\nabla}) + (eA_0\gamma^0 - e\mathbf{A} \cdot \boldsymbol{\gamma}) - m_0] \psi \\ &= \bar{\psi} [i\gamma^\mu \partial_\mu + e\gamma^\mu A_\mu - m_0] \psi \\ &= \bar{\psi} [i\gamma^\mu (\partial_\mu - ieA_\mu) - m_0] \psi \end{aligned} \quad (83)$$

In order to be convenient to study the system, we transform the system from Minkowski space-time to Euclidean spacetime:

$$S_{\text{TI}}^E = -iS_{\text{TI}}. \quad (84)$$

We rewrite t , A_0 and γ^j as following:

$$\left\{ \begin{aligned} t &\rightarrow -i\tau \\ A_0 &\rightarrow iA_0 \\ \gamma^j &\rightarrow i\gamma_j \\ m_0 &= -m_0(\cos \pi + i\gamma_5 \sin \pi) = -m_0 e^{i\pi\gamma_5} \end{aligned} \right. \quad (85)$$

So the Euclidean spacetime action of system is written as:

$$S_{\text{TI}}^E = \int d\tau d^3x \bar{\psi} [\gamma_\mu (\partial_\mu - ieA_\mu) - m_0 e^{i\pi\gamma_5}] \psi \quad (86)$$

Now we apply Fujikawa's method to the action. First, let's consider an infinitesimal chiral transformation defined by:

$$\begin{cases} \psi \rightarrow \psi' = e^{-\frac{i\pi\phi}{2}\gamma_5} \psi \\ \bar{\psi} \rightarrow \bar{\psi}' = \bar{\psi} e^{-\frac{i\pi\phi}{2}\gamma_5} \end{cases} \quad (87)$$

where $\phi \in [0, 1]$. Then the partition function Z is transformed as:

$$\begin{aligned} Z &= \int D[\psi, \bar{\psi}] e^{-S_{\text{TI}}^E[\psi, \bar{\psi}]} \\ \Rightarrow Z' &= \int D[\psi', \bar{\psi}'] e^{-S_{\text{TI}}^E[\psi', \bar{\psi}']} \end{aligned} \quad (88)$$

let's see the first term of $S_{\text{TI}}^{E'}$:

$$K = \int d\tau d^3x \bar{\psi} e^{-i\theta\gamma_5} [\gamma_\mu (\partial_\mu - ieA_\mu)] e^{-i\theta\gamma_5} \psi \quad (89)$$

where $\theta = \frac{\pi d\phi}{2}$. Because the anticommutation relation $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$ is satisfied, so:

$$e^{-i\theta\gamma_5} \gamma_\mu = \gamma_\mu e^{i\theta\gamma_5} \quad (90)$$

The Lagrangian of K can be written as:

$$\begin{aligned} \mathcal{L}_K &= \bar{\psi} \gamma_\mu e^{i\theta\gamma_5} (\partial_\mu - ieA_\mu) e^{-i\theta\gamma_5} \psi \\ &= \bar{\psi} \gamma_\mu (\partial_\mu - ieA_\mu) \psi \end{aligned} \quad (91)$$

Next, let's see the mass term of $S_{\text{TI}}^{E'}$:

$$\begin{aligned} \mathcal{L}_M &= -\bar{\psi} e^{-i\theta\gamma_5} m_0 e^{i\pi\gamma_5} e^{-i\theta\gamma_5} \psi \\ &= -m_0 \bar{\psi} e^{i\pi(1-d\phi)\gamma_5} \psi \end{aligned} \quad (92)$$

So the action in Euclidean space time is transformed as:

$$S_{\text{TI}}^{E'} = \int d\tau d^3x \bar{\psi} [\gamma_\mu (\partial_\mu - ieA_\mu) - m_0 e^{i\pi(1-d\phi)\gamma_5}] \psi \quad (93)$$

We should consider the transformation of $D[\psi', \bar{\psi}']$

$$D[\psi', \bar{\psi}'] = JD[\psi, \bar{\psi}] \quad (94)$$

The Jacobian J is written as:

$$\ln J = -i \int d\tau d^3x d\phi \frac{\pi e^2}{16\pi^2} \star F^{\mu\nu} F_{\mu\nu} \quad (95)$$

We integrate with respect to the variable ϕ from 0 to 1. Due to the invariance of the partition function, finally, we arrive at the following expression of S_{TI}^E :

$$S_{TI}^E = \int d\tau d^3x \bar{\psi} [\gamma_\mu (\partial_\mu - ieA_\mu) - m_0] \psi + i \int d\tau d^3x \frac{\pi e^2}{16\pi^2} \star F^{\mu\nu} F_{\mu\nu} \quad (96)$$

The first term is the action of a topological trivial insulator since the mass $-m_0$ is positive. The second term is the θ term whose θ equal to π in the imaginary time. Thus, we have demonstrated that for 3D topological insulators, the θ in θ term is π .

2.1 Topological magneto electric effect

we have seen that topological insulators are a class fascinating class of materials whose dynamic is characterised by the fact that $\theta = \pi$. we will see that if a topological insulator is placed in an electric or magnetic field, it will exhibit many novel physical effects.

First, we place a topological insulator in a uniform magnetic field directed along the z axis. Because θ undergoes a discontinuity at the surface of the topological insulator, charge will be induced on the surface:

$$\int \sigma \cdot d\mathbf{s} = \int \nabla \cdot \mathbf{E} dV = \int \frac{\alpha}{\pi} (\nabla\theta) \cdot \mathbf{B} dV. \quad (97)$$

At the upper interface, θ jumps from π to 0, whereas at the lower interface, θ jumps from 0 to π , so:

$$\sigma = \begin{cases} -\alpha B & \text{(upper interface)} \\ \alpha B & \text{(lower interface)} \end{cases} \quad (98)$$

Thus, the magnetic field induces negative charge on the upper interface and positive charge on the lower interface.

Alternatively, we can turn on an electric field lies tangential to the interface, we place a topological insulator in uniform electric field directed along the y axis. Similarly, because θ undergoes a discontinuity at the surface of the topological insulator, a surface current $\mathbf{K}$ will therefore be induced:

$$\int \mathbf{K} \cdot d\mathbf{l} = \int \nabla \times \mathbf{B} ds = - \int \frac{\alpha}{\pi} \nabla\theta \times \mathbf{E} ds \quad (99)$$

Simplistically, we only consider the surface current on upper interface, so

$$\begin{aligned} \mathbf{K} &= -\alpha E \hat{e}_x \\ &= -\frac{1}{2} \cdot \frac{e^2}{2\pi} E \hat{e}_x \end{aligned} \quad (100)$$

The creation of a two-dimensional current which lies perpendicular to an applied electric field is called the Hall effect the coefficient of proportionality is known as the Hall conductivity:

$$\sigma_{xy} = \frac{1}{2} \cdot \frac{e^2}{2\pi} \quad (101)$$

This is usually abbreviated to say that the interface of the topological insulator has Hall conductivity $1/2$. The general phenomenon in which electric fields induce magnetic fields and vice versa goes by the name of the topological magneto-electric effect.

3 Weyl semimetals

The Weyl semimetals have nondegenerate gapless linear dispersions around band touching points (Weyl nodes). The low-energy effective Hamiltonian around a Weyl node is written as:

$$H_W(\mathbf{k}) = Qv_F \mathbf{k} \cdot \boldsymbol{\sigma} \quad (102)$$

where $Q = \pm 1$ indicates the chirality, v_F is the Fermi velocity and σ_i are Pauli matrices. Because the sum of chiralities of Weyl nodes in a system must be zero, the simplest realization of a Weyl semimetal is one with two Weyl nodes of opposite chiralities. Note that the minimal number of Weyl nodes in Weyl semimetal with broken inversion symmetry is four, while it is two in Weyl semimetals with broken time-reversal symmetry.

Now, let's consider a 4×4 continuum model Hamiltonian for two-node Weyl semimetals with broken time-reversal symmetry:

$$H_0(\mathbf{k}) = v_F(\tau_z \otimes \mathbf{k} \cdot \boldsymbol{\sigma} + \Delta\tau_x \otimes \mathbb{I} + \mathbb{I} \otimes \mathbf{b} \cdot \boldsymbol{\sigma}) \quad (103)$$

where τ_i and σ_i are Pauli matrices for Weyl-node and spin degrees of freedom respectively and Δ is the mass of 3D Dirac fermions. The term $\mathbf{b} \cdot \boldsymbol{\sigma}$ represents a magnetic interaction such as exchange interaction between conduction electrons and magnetic impurities or the Zeeman coupling with an external magnetic field. Note the Hamiltonian with $\mathbf{b} = 0$ describes a topological or normal insulator depending on the sign of Δ (mass). Without loss of generality, we may set $\mathbf{b} = (0, 0, b)$.

Now let's find the location of Weyl nodes. The Hamiltonian of Weyl nodes can be written as:

$$\begin{aligned} H &= v_F \tau_z (k_x \sigma_x + k_y \sigma_y + k_z \sigma_z) + \Delta \tau_x \otimes \mathbb{I} + b \mathbb{I} \otimes \sigma_z \\ &= \underbrace{v_F \tau_z \otimes (k_x \sigma_x + k_y \sigma_y)}_{H_\perp} + \underbrace{v_F \tau_z \otimes k_z \sigma_z + \Delta \tau_x \otimes \mathbb{I} + b \mathbb{I} \otimes \sigma_z}_{H_\parallel} \end{aligned} \quad (104)$$

Because $H_\perp$ and $H_\parallel$ anticommute, so we can write down:

$$\begin{aligned} H^2 &= H_\perp^2 + H_\parallel^2 + \{H_\perp, H_\parallel\} \\ &= H_\perp^2 + H_\parallel^2 \end{aligned} \quad (105)$$

This leads to the explicit energy spectrum:

$$E^2 = v_F^2 (k_x^2 + k_y^2) + E_\parallel^2 \quad (106)$$

and $H_\parallel$ can be split into two 2×2 blocks:

$$\begin{aligned} h_{eff} &= s(k_z \tau_z + b) + \Delta \tau_x \\ &= s \mathbf{k}_z \cdot \boldsymbol{\tau}_z + \Delta \tau_x + s b \mathbb{I} \end{aligned} \quad (107)$$

where $s = \pm 1$. So the energy spectrum of h_{eff} is given by:

$$E_{\parallel} = sb \pm \sqrt{k_z^2 + \Delta^2} \quad (108)$$

So the Weyl node location satisfies:

$$\begin{cases} k_x^2 + k_y^2 = 0 \\ sb \pm \sqrt{v_F^2 k_z^2 + \Delta^2} = 0 \end{cases} \quad (109)$$

Thus, the coordinates of the Weyl nodes are:

$$(k_x, k_y, k_z) = (0, 0, \pm \frac{\sqrt{b^2 - \Delta^2}}{v_F}) \quad (110)$$

which requires $b^2 > \Delta^2$.

And then we derive the θ term of Weyl semimetals from the microscopic four-band model. In order to describe a more generic Weyl semimetal, we add the term $\mu_5 \tau_z$ to the Hamiltonian which generates a chemical potential difference $2\mu_5$ between the two Weyl nodes. We also set $\Delta = 0$ for simplicity. So the Hamiltonian can be written as:

$$H'_0 = v_F(\tau_z \otimes \boldsymbol{\sigma} \cdot \mathbf{k} + \mathbb{I} \otimes \boldsymbol{\sigma} \cdot \mathbf{b}) - \mu_5 \tau_z \otimes \mathbb{I} \quad (111)$$

We rewrite the Hamiltonian with Gamma matrices.

$$\begin{cases} \gamma^0 = \tau_x \otimes \mathbb{I} \\ \gamma^i = -i\tau_y \otimes \sigma_i \\ \gamma^5 = \tau_z \otimes \mathbb{I} \\ b_\mu = (\mu_5, -\mathbf{b}) \end{cases} \quad (112)$$

and we set $v_F \approx c = 1$, so:

$$H = -\gamma^0 \gamma^i k_i + \gamma^0 \gamma^i \gamma^5 b_i - \mu_5 \gamma^5 \quad (113)$$

The action of the system in the presence of external electric and magnetic fields with the four potential $A_\mu = (A_0, -\mathbf{A})$ is given by

$$\begin{aligned} S &= \int dt d^3x \psi^\dagger [i(\partial_t - ieA_0) - H'_0(\mathbf{k} + e\mathbf{A})] \psi \\ &= \int dt d^3x \bar{\psi} \gamma^0 [i\partial_t + i\gamma^0 \gamma^i \partial_i + eA_0 - e\gamma^0 \gamma^i A_i - \gamma^0 \gamma^i \gamma^5 b_i + \mu_5 \gamma^5] \psi \\ &= \int dt d^3x \bar{\psi} i [\gamma^0 \partial_0 + \gamma^i \partial_i - ie(\gamma^0 A_0 - \gamma^i A_i) - i(\gamma^0 \mu_5 - \gamma^i b_i) \gamma^5] \psi \\ &= \int dt d^3x \bar{\psi} i (\gamma^\mu \partial_\mu - ie\gamma^\mu A_\mu - i\gamma^\mu b_\mu \gamma^5) \psi \\ &= \int dt d^3x \bar{\psi} i \gamma^\mu (\partial_\mu - ieA_\mu - ib_\mu \gamma^5) \psi \end{aligned} \quad (114)$$

where $e > 0$, ψ is a four-component spinor, $\bar{\psi} = \psi^\dagger \gamma^0$.

Now we apply Fujikawa's method to the action. Performing an infinitesimal gauge transformation for infinite times such that:

$$\begin{cases} \psi \rightarrow \psi' = e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \psi \\ \bar{\psi} \rightarrow \bar{\psi}' = \bar{\psi} e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \end{cases} \quad (115)$$

with $\theta(\mathbf{r}, t) = 2x^\mu b_\mu = 2(\mathbf{b} \cdot \mathbf{r} - \mu_5 t)$ and $\phi \in [0, 1]$, then the partition function Z is transformed as:

$$\begin{aligned} Z &= \int D[\psi, \bar{\psi}] e^{iS[\psi, \bar{\psi}]} \\ \Rightarrow Z' &= \int D[\psi', \bar{\psi}'] e^{iS[\psi', \bar{\psi}']} \end{aligned} \quad (116)$$

Because the anticommutation relation $\{\gamma^\mu, \gamma^5\} = 0$ is satisfied, so:

$$e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \gamma^\mu = \gamma^\mu e^{\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \quad (117)$$

The Lagrangian of $S[\psi', \bar{\psi}']$ can be written as:

$$\begin{aligned} \mathcal{L} &= \bar{\psi} e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \cdot i\gamma^\mu (\partial_\mu - ieA_\mu - ib_\mu \gamma^5) e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}} \psi \\ &= \bar{\psi} i\gamma^\mu (\partial_\mu - ieA_\mu) \psi + \bar{\psi} i\gamma^\mu (\partial_\mu e^{-\frac{i\theta(\mathbf{r},t)\gamma^5 d\phi}{2}}) \psi - \bar{\psi} i\gamma^\mu (-ib_\mu \gamma^5) \psi \\ &= \bar{\psi} i\gamma^\mu (\partial_\mu - ieA_\mu) \psi - \bar{\psi} i\gamma^\mu (1 - d\phi) ib_\mu \gamma^5 \psi \end{aligned} \quad (118)$$

And the Jacobian J of transformation of $D[\psi', \bar{\psi}'] = JD[\psi, \bar{\psi}]$ is given by:

$$\ln J = i d\phi \int dtd^3x \frac{e^2}{16\pi^2} \cdot \theta(\mathbf{r}, t) \cdot {}^*F^{\mu\nu} F_{\mu\nu} \quad (119)$$

We integrate with respect to the variable ϕ from 0 to 1. Due to the invariance of the partition function, we arrive at the following expression of S :

$$S = \int dtd^3x \bar{\psi} i\gamma^\mu (\partial_\mu - ieA_\mu) \psi + \frac{e^2}{16\pi^2} \int dtd^3x \theta(\mathbf{r}, t) {}^*F^{\mu\nu} F_{\mu\nu} \quad (120)$$

where the first term represents the trivial action of massless Dirac fermions and the second term is the θ term with $\theta(\mathbf{r}, t) = 2(\mathbf{b} \cdot \mathbf{r} - \mu_5 t)$.

3.1 Anomalous Hall effect and negative magnetoresistance

Let's consider the consequences of the presence of a θ term in Weyl semimetals. As we have also seen in the case of insulators with a dynamical axion field, an electric current is induced in the presence of a θ term. The induced electric current density and charge density are given by:

$$\begin{cases} \mathbf{j}(\mathbf{r}, t) = -\frac{\alpha}{\pi} (\dot{\theta} \cdot \mathbf{B} + \nabla\theta \times \mathbf{E}) \\ \rho(\mathbf{r}, t) = \frac{\alpha}{\pi} (\nabla\theta) \cdot \mathbf{B} \end{cases} \quad (121)$$

In the present case of $\theta(\mathbf{r}, t) = 2(\mathbf{b} \cdot \mathbf{r} - \mu_5 t)$, we readily obtain a static current of the form:

$$\mathbf{j} = \frac{2\alpha}{\pi}(\mu_5 \mathbf{B} - \mathbf{b} \times \mathbf{E}) \quad (122)$$

The second term is the anomalous Hall effect. Let's consider a 2D plane which is perpendicular to $\mathbf{b}$. And we set $\mathbf{b} = (0, 0, b)$, which gives the 3D Hall conductivity as:

$$\sigma_{xy}^{3D} = \frac{2\alpha b}{\pi} \quad (123)$$

The first term is the chiral magnetic effect. The chiral magnetic effect indicates that a direct current is generated along a static magnetic field even in the absence of electric fields, when there exists a chemical potential difference $\delta\mu = 2\mu_5$ between the two Weyl nodes. However, the existence of the static chiral magnetic effect is ruled out in crystalline solid. But the chiral magnetic effect can be realized under nonequilibrium circumstance when the system is driven from equilibrium by the combined effect of electric and magnetic fields. In this case of Weyl semimetal potential difference can be generated by the chiral anomaly. The chiral anomaly in weyl semimetals is referred to as the electron number nonconservation in a given weyl cone under parallel electric and magnetic fields in which the rate of pumping of electrons is given by:

$$\frac{\partial N_i}{\partial t} = Q_i \frac{\alpha}{\pi} \mathbf{E} \cdot \mathbf{B} \quad (124)$$

where i is a valley (Weyl node) index and

$$Q_i = \int \frac{d^3k}{2\pi} \cdot \frac{\partial f_0(\mathcal{E}_k^m)}{\partial \mathcal{E}_k^m} \cdot \mathbf{v}_k^m \cdot \mathbf{\Omega}_k^m \quad (125)$$

is the chirality of the valley. Here, $\mathcal{E}_k^m$ is the energy of Bloch electrons with momentum $\mathbf{k}$ in band m in a given valley i , $f_0(\mathcal{E}_k^m)$ is the Fermi distribution function. $\mathbf{v}_k^m$ is the group velocity and $\mathbf{\Omega}_k^m$ is the Berry curvature. So the difference of the total electron number between the weyl nodes leads to the difference of the chemical potential between weyl nodes.

A phenomenon manifested by the chiral anomaly is a negative magnetoresistance:

$$\sigma_{zz} = \frac{\alpha}{\pi} \frac{(eB_z)^2 v_F^3}{\mu^2} \tau_{\text{inter}} \quad (126)$$

where μ is the equilibrium chemical potential and τ_{inter} is the intervalley scattering time. This unusual magnetoconductivity holds in the low-field limit $B_z \rightarrow 0$. So the expressing of $\mathbf{j}$ is understood as coming from

$$\mathbf{j} \propto (\mathbf{E} \cdot \mathbf{B} \tau_{\text{inter}}) \cdot \mathbf{B} \quad (127)$$

which indicates that it is a consequence of the chiral magnetic effect.